A3-PS003-TC337

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

Category: Smart Cities

Team Name: ALPHA_2.0

AIORI Hackathon 3

Problem Description

Accurate and efficient traffic-flow prediction is essential for smart city traffic management, but current approaches face key challenges:

High Data Transmission Cost

Centralized forecasting requires continuous streaming of raw data from edge cameras and roadside sensors, consuming high cellular bandwidth.

Privacy Concerns

Centralizing travel logs can concentrate sensitive movement data in a single cloud server, raising privacy risks.

Non-Identical Traffic Patterns

Different intersections and regions have diverse and non-iid traffic patterns, making it difficult for a single centralized model to generalize well.

Operational Constraints

Real-world factors such as signal halts, congestion, maintenance blocks, and variable speed limits introduce dynamic changes in traffic flow, which are not well captured by static models.

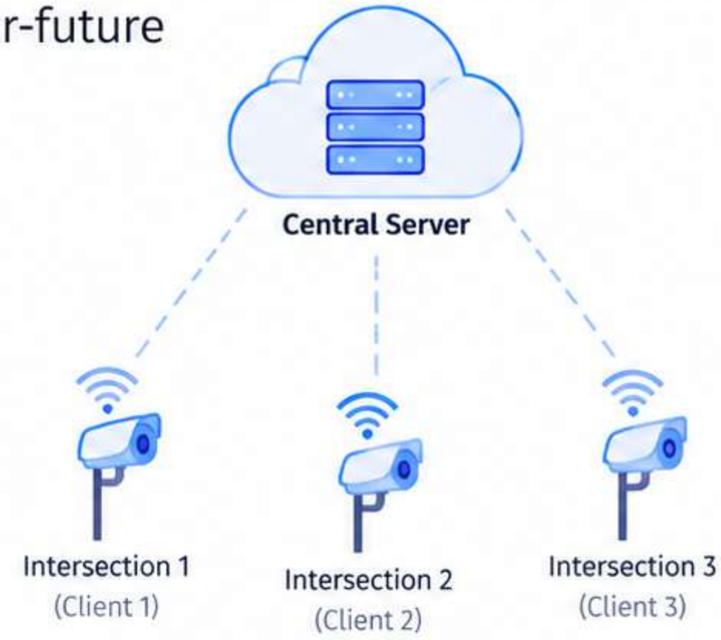


Background

Understanding traffic dynamics and distributed learning is essential to the proposed optimization.

Traffic flow changes over time and differs between road segments. Forecasting models use historical sensor measurements to learn recurring patterns and predict near-future traffic conditions.

- **Centralized learning** combines sensor data at a central server, which can increase raw-data transfer requirements.
- **Federated Learning** trains models locally at simulated intersections and shares model updates with a coordinating server.
- **FedAvg** combines client updates; MAE, RMSE, convergence, and transmitted bytes help compare prediction quality and communication cost.



Literature & RFC Review

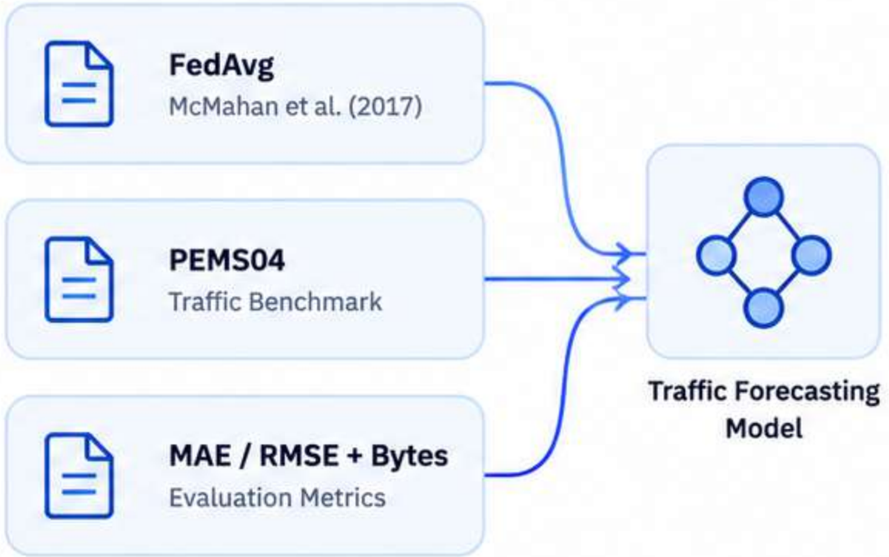
Existing research provides the foundations for federated optimization and traffic forecasting; our study evaluates the communication trade-off.

- **Federated Averaging (McMahan et al., 2017)**
  FedAvg combines locally trained model updates into a shared global model, without collecting clients' raw training examples at the server.

- **PEMS04 Traffic Benchmark**
  A public traffic dataset with 307 sensors and five-minute observations, suitable for evaluating traffic-flow forecasting.

- **Evaluation Gap Addressed by FedTraffic**
  Compare centralized and FedAvg models under IID and non-IID client distributions, measuring MAE, RMSE, convergence, and transmitted model bytes.

Research → Evaluation



Proposed Solution

A federated forecasting simulator compares prediction quality with communication cost.

FedTraffic trains a compact traffic-flow model using PEMS04 data and compares centralized learning with FedAvg across simulated intersections.



Forecast Accuracy
Test MAE and RMSE

Communication Cost
Upload + download bytes

Convergence
Validation error by round

Raw training examples remain local in the federated simulation; model updates are exchanged.

Optimization Proposed

The design improves evaluation fairness, communication tracking, and reproducibility.

Area	Baseline Approach	Proposed Optimization
Data handling	Uncontrolled or random splits	Chronological 70/10/20 train–validation–test split; no window crosses split boundaries
Client simulation	One data distribution	3–5 virtual clients; compare IID and sensor-based non-IID partitions
Model training	Centralized model only	Same compact GRU for centralized and FedAvg experiments
Communication	Cost not explicitly measured	Count serialized model download + upload bytes per round and cumulatively
Evaluation	Single-run metrics	Test MAE/RMSE, convergence tracking, and results across seeds 42, 123, 2026

Outcome: a reproducible, evidence-based comparison of forecast quality versus communication cost.

Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.

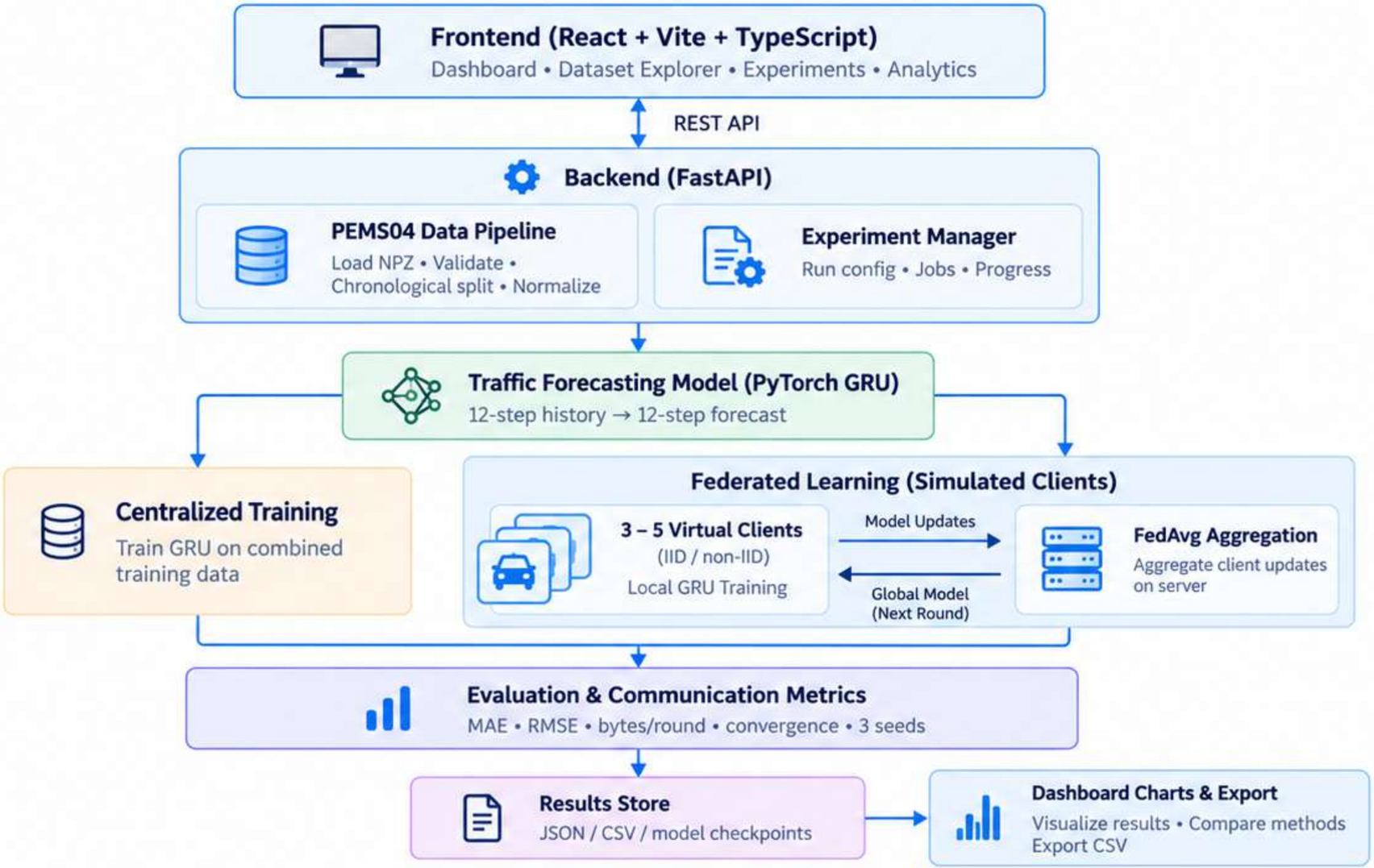


Figure: FedTraffic System Architecture

Data Flow Diagram

From PEMS04 traffiic observations to reproducible model and communication evaluation



Raw training examples stay local in the federated simulation; model updates are exchanged.

Deployment Timeline

Milestones from dataset preparation to a validated, demonstrable prototype.

1

Dataset & Setup

Load and validate PEMS04; configure frontend and FastAPI backend.

2

Forecasting Model

Preprocess chronological splits; train the compact GRU baseline.

3

Federated Simulator

Create 3–5 virtual clients; implement IID/non-IID partitions and FedAvg.

4

Evaluation

Measure test MAE/RMSE, convergence, and upload/download bytes.

5

Validation & Demo

Repeat seeds 42, 123, 2026; test APIs and finalize results.

Deliverable checkpoints

Working website + API
Frontend dashboard and backend services running

Reproducible experiment results
Centralized and federated runs with 3 seeds

Charts, CSV export + report
Performance, communication metrics and comparison results

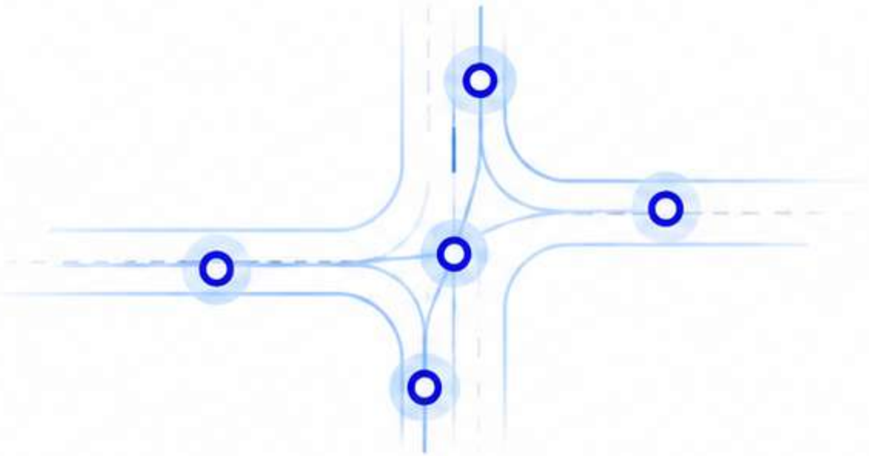
Prototype scope: historical data and simulated clients; no live-camera deployment.

Bibliography I

[1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas. "Communication-Efficient Learning of Deep Networks from Decentralized Data." In: *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, PMLR 54 (2017), pp. 1273–1282.

[2] D. Han et al. "A New Traffic Prediction Method Based on Spatial-Temporal Feature Learning." Use the PeMSD4 / PEMS04 traffic benchmark repository for the project data source: Davidham3, "ASTGCN — PEMS04 Dataset," GitHub repository, https://github.com/Davidham3/ASTGCN/tree/master/data/PEMS04.

[3] K. Cho et al. "Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation." In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP 2014)* (2014), pp. 1724–1734.



Thank You!

Questions • Discussion